Yichen Wang, M.Sc. in Computer Science

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SUMMARY

Frontend Engineer with 6+ years of experience building production-grade React and TypeScript applications backed by AWS-based APIs. Experienced in designing scalable, API-driven web platforms with strong focus on structured data modeling, reliability, and automated testing. Proven ability to translate backend contracts into reusable, accessible UI systems while maintaining production stability. Passionate about engineering rigor, performance optimization, and clean architecture.

PROFESSIONAL EXPERIENCE

IBM

Senior Software Engineer *02/2022 - PRESENT*

- Delivered 500+ enhancements across three major production releases of a large public-facing web platform.
- Led frontend modernization effort, migrating from legacy React architecture to a modular Svelte + TypeScript system to improve performance and maintainability.
- Partnered closely with backend engineers to validate API contracts, enforce schema correctness, and ensure consistent frontend–backend integration.
- Authored 300+ automated tests (Jest + Playwright), increasing coverage from 40% to 85% and reducing production incidents by 60%.
- Acted as technical lead during release cycles, coordinating DEV → UAT → PROD deployments and resolving production issues.
- Worked cross-functionally with UX and product to deliver accessible, responsive, and user-centered components.

University at Buffalo

Full Stack Developer *11/2019 - 07/2021*

- Built structured React-based research applications with dynamic filtering, forms, and dashboards.
- Designed reusable UI components aligned with accessibility standards.
- Developed backend utilities for structured data export and validation (PHP → CSV pipelines).
- Collaborated with researchers to ensure system reliability and schema accuracy.

INDEPENDENT PROJECT EXPERIENCE

Cross-Platform Social App (React Native + AWS + AI integrations)

- Architected and deployed a scalable social platform with React Native frontend and Node.js backend hosted on AWS (EC2, S3, DynamoDB).
- Designed API-driven multilingual content pipelines using asynchronous AWS Translate workflows, persisting structured language-specific representations.
- Integrated OpenAI APIs to generate structured summaries and enhance content discoverability.
- Implemented custom authentication system (email + OTP via SendGrid & Twilio) with secure account lifecycle management.
- Built automated content moderation workflows using AWS Rekognition to maintain platform integrity.
- Designed reusable, declarative UI systems supporting nested threads and real-time state reconciliation.

Lost & Found Pet Platform (React + Supabase + Map integrations)

- Built a geospatial React application supporting real-time data updates and map-driven search workflows.
- Designed scalable data models enabling efficient query performance and structured rendering.
- Implemented client-side caching strategies to reduce latency and improve responsiveness.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, SQL
- **Frontend:** React.js, Next.js, Svelte, React Native, Tailwind CSS, Responsive Design
- **API & Platform:** RESTful API Design, Structured Data Modeling, Supabase (Postgres, Auth)
- **Cloud & Infra:** AWS (EC2, S3, Lambda, DynamoDB, API Gateway), Docker
- **Testing:** Jest (Unit/Integration), Playwright (E2E), CI/CD (GitHub Actions)
- **Tooling:** Git, Vercel, Render

EDUCATION

University at Buffalo — Buffalo, NY

- M.S. in Computer Science and Engineering — May 2020
- B.S. in Computer Science, Magna Cum Laude — Dec 2018